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Effect of CO₂-Water-Rock Reaction on Rock Properties Changing in a High Salinity Low Permeability Reservoir

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Abstract

CO₂ utilization for enhanced oil recovery and sequestration is the most efficiency way to achieve carbon neutrality. Especially for those low permeability oil reservoirs with poor water injectability, CO₂ flooding presents the great application potential. CO₂ flooding performance may be severely affected by the precipitation and erosion which produced by the reaction of CO₂-water-rock system in the high salinity formation water reservoirs.

This work visualized the processes and impacts of inorganic salt precipitation using the high temperature high-pressure visible reactor during CO₂ injection. The effect of its interaction with mineral dissolution and particle transport on the porosity and permeability changing were investigated via the nuclear magnetic resonance, scanning electron microscopy and core flowing experiments. CO₂ flowing in the dual cores were conducted and the production amounts of CO₂ in each core were recorded for the heterogeneous changes analysis during the CO₂ flooding process.

Results show that the precipitation of inorganic salt was mainly composed by the carbonate minerals. It uniformly adsorbed on the surface of the pore throat and resulted in an overall decrease in the radius of the pore throat. The CO₂ erosion mainly occurred in macro pores and effectively enlarged the porosity of cores up to 1.9% at most. With the increasing of the core permeability, the effect of CO₂ dissolution was strengthened while the effects of inorganic salt precipitation and particle transport were weakened. After multiple rounds of CO₂ injection in dual cores, the permeability of high-permeability core varied from 1.51mD to 2.24mD, while the permeability of low-permeability core changed from 0.23mD to 0.26mD. The permeability difference increased from 6.5 to 8.7 and gas channel was formed in the high-permeability core. This work revealed the porosity and permeability changing rule upon the interaction of salt precipitation,

mineral dissolution and particle transport during the CO₂ injection, which could provide the guidance for the CO₂ flooding design.

Introduction

Injecting CO₂ into underground is a win-win approach for the oil production and CO₂ sequestration. To those low permeability reservoirs whose failed in the water injection, CO₂ flooding presents a huge application potential due to its good injectability, easy miscible with oil and high oil recovery (Lu et al., 2021; Liu et al., 2022; Song et al., 2023; Li et al., 2023). However, CO₂ could react with the formation water and mineral of the rocks during its injection, which may alter rock surface micro-morphology and pore structure. These reactions including the inorganic salt precipitation, mineral dissolution and particle migration could significantly affect CO₂ flooding performance and CO₂ storage safety via changing the reservoir properties. Therefore, the effect of CO₂-water-rock reaction on the changing of core properties is crucial for the application and promotion of CO₂ flooding.

CO₂ dissolved in the formation water since it was injected into the reservoir. The dissolved CO₂ could form H₂CO₃ and then ionize to HCO₃⁻ and CO₃²⁻. These ions combined with Ca²⁺ and Mg²⁺ in the brine will produce CaCO₃ and MgCO₃ precipitates. The pressure, temperature, pH value of formation water and rock mineral component jointly influence this process (Zhu et al., 2011). Nasser et al. (2015) and Chen et al. (2004) observed the growth of CaCO₃ crystals in aqueous solutions over time and found that the growth rate was significantly affected by the supersaturation of calcium-containing solutions and the concentration of Ca²⁺. In the process of oil production, people focus on the effect of inorganic salt precipitation on the permeability and porosity changing. Yuan et al. (2021) found the amount the precipitation was positive with the pressure and this phenomenon was more obviously nearly the production well. Zeidouni et al. (2009) and Sbail et al. (2011) both believed the inorganic salt precipitation could decrease the permeability via the numerical simulation. It seems that there is no wonder that inorganic salt precipitation narrowed the fluid flow channel. However, the dissolution caused by CO₂ injection could not be neglected.

In addition to reacting with ions in the water to form inorganic precipitates, CO₂ dissolved in formation water can also react with rocks and potentially producing new minerals like kaolinite and silica (Busenberg and Plummer, 2011; Azin et al., 2015). Yu et al. (2012) found the carbonate minerals dissolved more than feldspar, while the calcite dissolving the most via the scanning electron microscope (SEM) and whole-rock X-ray diffraction analyze. Fischer et al. (2020) confirmed silicate minerals and anhydrite of the sandstone could both present a noticeable dissolution phenomenon. Tutolo et al. (2015) found the primary feldspars and precipitate secondary aluminum minerals could be dissolved as the acidity incremental with CO₂ injection. This process could change the permeability, porosity and wettability of the cores.

There is no clear conclusion how does mineral dissolution affect the rock properties. Several researchers (Luquot et al, 2012; Garcia-Rios et al., 2015) have observed significant dissolution of carbonate minerals in experiments. The dissolution rate decreased with increment of temperature and pressure, while the temperature exerted a more dominant influence than pressure (Bacci et al., 2010). Dissolution-induced pores and micro-fractures significantly enhanced core properties by the pore connectivity improvement and permeability increment (Li et al., 2019). However, this process will compromise the rock's tensile and compressive strength. Mineral dissolution also creates secondary pores. Othman et al. (2019) noted that pore - throat blockage caused by particle migration, secondary mineral precipitation and clay particle release could significantly reduce permeability, while having minimal impact on porosity. Al-Yaseri et al. (2017) found that small pore - throat radii and scattered clay particles were more likely to cause a reduction in the permeability. Yu et al. (2018) used micro-CT scans to observe clay particles mobilized at the core inlet and accumulated at the outlet, it could lead a 50% reduction in core permeability. Wang et al. (2020, 2021) believed that asphaltene precipitation was the primary reason for the permeability changing, although CO₂-brine-rock interactions also played a significant role. Some studies revealed that porosity and permeability

evolved dynamically action. Xiao et al. (2017) found porosity initially increased and then decreased in the CO₂-water-CaCO₃ soaking experiments. The dissolution was dominant in the early stages and then the kaolinite precipitation caused by the potassium feldspar reaction would lead to a decline in the porosity. Reynolds et al. (2014) conducted two-phase flow experiment in sandstone and found that permeability initially decreased but later recovered. He attributed it to the clay mineral blockage caused by dissolution and the acidic effect of CO₂. Xu et al. (2023) simulated the geo-chemical reactions in the reservoir following acid injection using the TOUGHREACT soft-ware. They found that compared to the far-well zone, the permeability increased significantly in the near well zone at the early stage, but it decreased due to the mineral precipitation at the later stage.

The above work indicated CO₂-work-rock reaction was still a hot issue in the CO₂ flooding and sequestration process. But there were even some contradictions in the reservoir permeability changing under the CO₂-work-rock reaction. This work preferred to explore the characteristics of inorganic salt precipitation generated by the interaction of formation water with CO₂. The effect of mineral dissolution during the CO₂ injection on the porosity and permeability changing were also investigated via the nuclear magnetic resonance, scanning electron microscopy and core flowing experiments. The reservoir properties changing law reveled in this work may provide time-varying permeability insight for the efficient CO₂ flooding in the low permeability heterogeneous sandstone reservoirs.

Experimental Materials and Methods

Experimental Materials and Instruments

The formation water and core samples used in the experiments were obtained from an oil field in China. The formation water is of the CaCl₂ type and has a salinity of 15145.7 mg/L. The average porosity of the reservoir block is 12.7%, and the average permeability is 1.62 mD. The pilot was characterized as a medium-porosity and low-to-ultra-low-permeability reservoir. The core samples with a diameter of 2.5 cm and a length of 5 - 7 cm were used in this work. The gas used in the experiments is CO₂ with a purity of 99.9%.

Inorganic Salt Precipitation Visualization Experiment

A high-temperature and high-pressure visual reactor was used to mimic the downhole conditions. Glass beads were used to simulate the pore-throat structure. The experiment aimed to investigate the interaction between CO₂ and formation water in the pore structure and the visualized the processes of inorganic salt precipitation. First, formation water and glass beads were placed in the reactor, heated to 110°C and aged. Then, CO₂ was injected to reach the pressures of 30 MPa. A high-definition camera was used to record the precipitation process, and a high-magnification microscope was used to observed the inorganic salt morphology on the glass beads' surfaces to analyze the precipitation characteristics.

CO₂ Dissolution Experiment Using Reservoir Cores

The static core dissolution experiment under high-temperature and high-pressure conditions can effectively presented the CO₂ dissolution process in reservoirs. It can help us to understand the variation pattern of rock physical properties. Cores with different permeabilities varied from 2mD to 50mD from same oilfield were selected., these cores were used for a 28-day of CO₂ dissolution experiment under the high-temperature and high-pressure conditions after they were saturated with brine. The experimental procedure was shown in Fig. 1. The porosity and permeability of the cores before and after the reaction were measured. Then, a scanning electron microscope was used to observe the micro-morphological changes on the pore surfaces of the cores. The changes in pore-throat sizes within the cores were analyzed using nuclear magnetic resonance (NMR) T_2 spectra to examine the effect of permeability on the dissolution process.

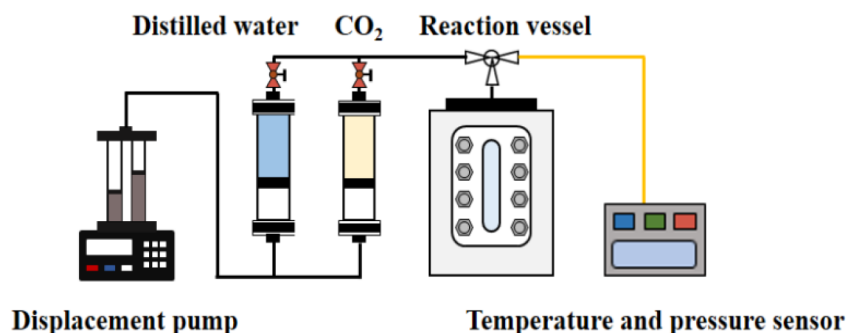


Figure 1—Schematic of the visualization experiment for the investigation of inorganic salt precipitation

Multi-round CO₂ Injection Experiment

The inorganic salt precipitation, mineral dissolution and particle migration were considered together in this section to understand the property changes of reservoir during CO₂ flooding. Two natural cores with the initial permeability of 1.51mD and 0.23mD were in parallel for a multi-stage CO₂ injection experiment. The experimental schematic was shown in Fig. 2. To rule out the impact of mineral salt segregation on reservoir properties, CO₂ saturated with water was injected slowly (0.05 ml/min). Each stage lasted 24 hours, then took out the core followed by permeability and NMR T_2 spectrum measurements. Nuclear magnetic resonance imaging was scanned before and after reactions for the analysis of the effects of time-dependent CO₂-water-rock interactions on the permeability and pore-size changes.

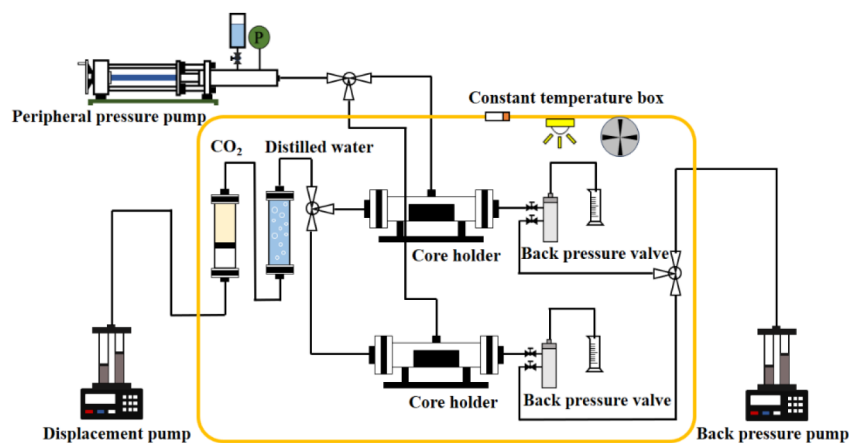


Figure 2—Schematic of Multi-round CO₂ Injection Experiment Process Flow Chart

Results and Discussion

Inorganic Salt Precipitation Process

The inorganic salt precipitation processes between CO₂ and brine were shown in Fig. 3. A large amount of white precipitate will soon appear at the gas-water interface after CO₂ injection. Since the density of the precipitate was greater than that of water, it will rapidly move downward. With the continuous injection of CO₂, the amount of precipitate will keep increasing until it covers the entire aqueous phase space. Then most of precipitates will be dissolved. This process was shown in Fig. 3(a). The inorganic precipitation process was almost the same when there were glass beads in the visual reactor. The precipitation formation and its dissolution were also observed. The red-framed parts in Fig. 3(b) were precipitation. Due to the limitation of pore size, capillary forces suppressed convection, slow down the CO₂ diffusion in the water, which make precipitation formation slow (Nagatsu et al. 2014). After the excess CO₂ injection, precipitation dissolved and residual precipitates evenly distributed in the glass beads space.

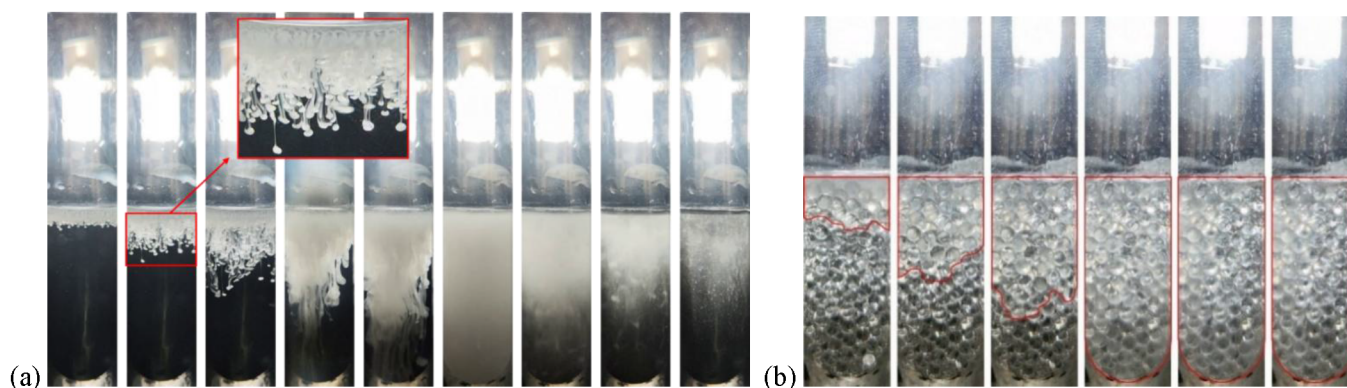


Figure 3—The process of inorganic salt precipitation and its dissolution during the CO₂ injection: (a) in bulk fluids; (b) in glass beads

Fig. 4 compared the glass bead surface before and after inorganic salt precipitation. The beads surface was covered by a frosted-texture precipitate, which uniformly and completely coated the beads after CO₂-brine reaction. It indicated that injecting CO₂ into the formation could cause the uniform crystallization. The crystallized precipitation may evenly cover the pore surfaces and reduce the pore-throat size for both high permeability cores and low permeability cores.

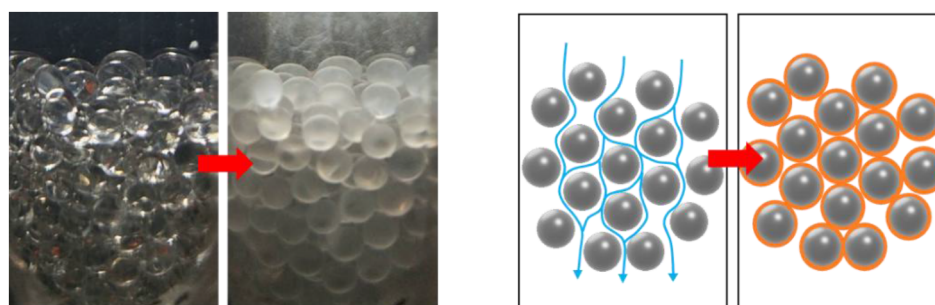


Figure 4—Comparison diagram of glass beads before and after the precipitation of inorganic salts

Images of glass beads under a high-magnification microscope in Fig.5 were large fascicular clusters and small rhombic crystals. They were both the typical carbonates, which indicated the final precipitates from the reaction of CO₂ and formation water were insoluble carbonates, such as CaCO₃ and MgCO₃. The ion composition and content were fixed in the experimental simulated water. But in the real reservoirs, metal ions produced by the mineral dissolution should be continuously supplied into the formation water. It would enable the precipitation process continuously proceed, which may lead to the permeability reduction and fluid flow channel blockage in the small pore-throats.

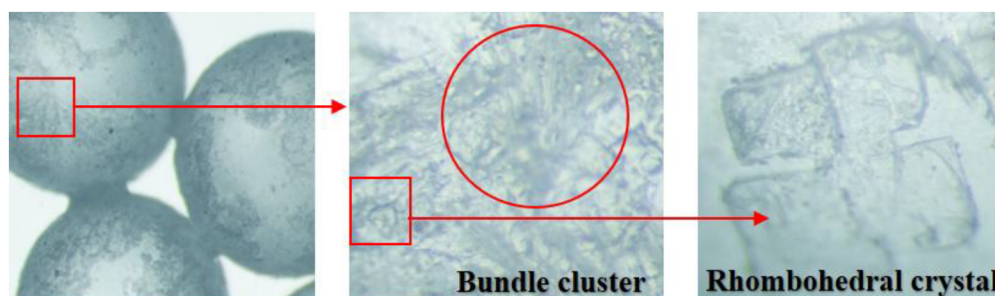


Figure 5—The images of inorganic salts precipitation on the glass beads surface

Changes in Pore Micro-Morphology

The SEM images of core after CO₂-water-rock reactions in Fig. 6 revealed two permeability-altering phenomena caused by dissolution. Feldspar minerals showed obvious dissolution phenomena after the brine-rock reaction, which result in the formation of numerous dissolution pits and book-like kaolinite. The accumulation of Kaolinite in pores will damage permeability. Moreover, the pore surfaces also produced free fine particles that could move along with the injected fluids. It could clog small throats and cause a significant permeability decline. Clay particle migration and accumulation were the main reasons for reservoir permeability damage. The water-rock reaction can also enhance core properties. The porosity would increase when the dissolution pits developed further into the vugular pore. Long-term dissolution could weaken the rock's tensile and compressive strength, making the formation of secondary fractures under the high pressure. These micro-fractures improved pore connectivity and enhanced core permeability.

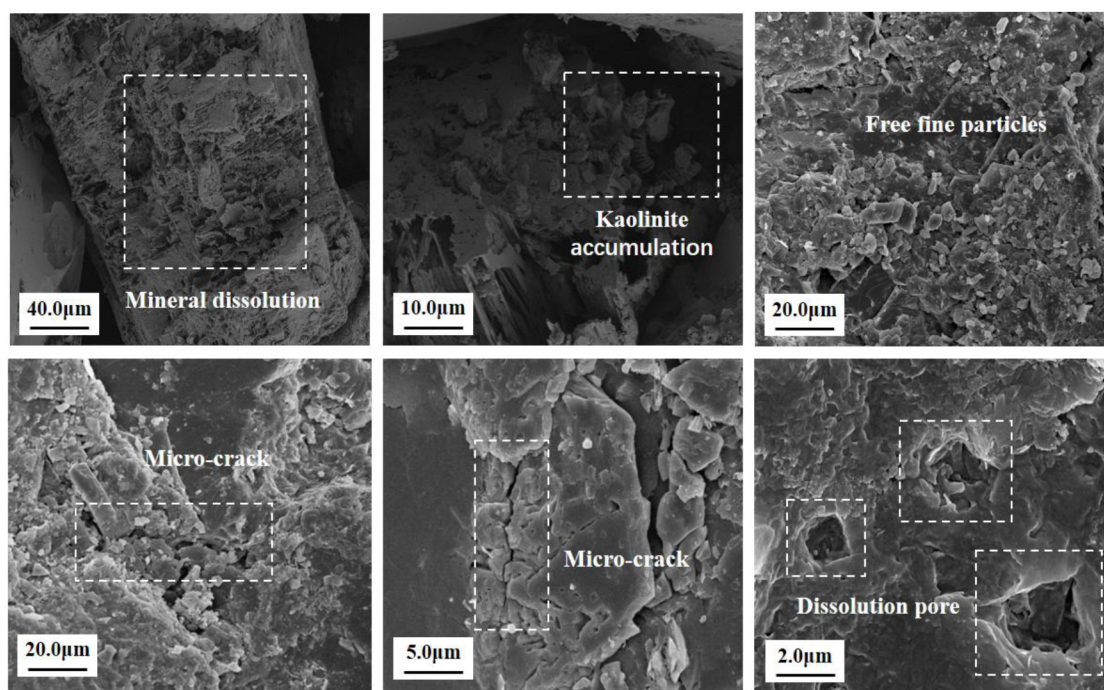


Figure 6—SEM images of the core surfaces after CO₂-water-rock reactions

Variation Laws of Core Physical Properties

The permeability and porosity changing of a core with an initial gas permeability of 2.45mD before and after the reaction were shown in Fig. 7(a). Its porosity increased from 17.25% to 17.3%, rising by 0.29% under the erosion of CO₂, while the water permeability decreased from 0.132mD to 0.109mD, reducing by 17.42%. The increment of porosity was rooted from the dissolution of certain soluble minerals due to the erosion effect. This process also released a multitude of fine particles. When these particles migrate within the core, they can block some pore throats and decrease the permeability. The NMR T_2 spectrum in Fig. 7(b) can present the distribution of pore and throat radii within the core. The NMR T_2 spectrum of low permeability exhibits a bimodal pattern. The increase of the signal intensity in the left peak and the decrease in the right peak indicated some throats had been blocked by particles and transformed into the throats with the smaller radii.

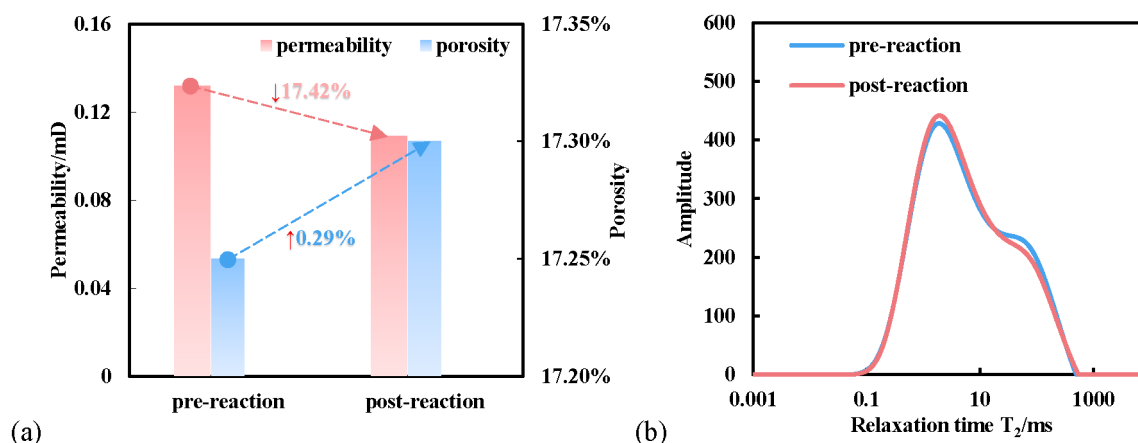


Figure 7—Properties change of a core with initial gas permeability of 2.45mD: (a) porosity and permeability; (b) NMR T_2 spectra

The effect of initial permeability on properties changing under the CO_2 dissolution were compared using different cores. Their permeability, porosity and NMR T_2 spectrum were shown in Fig. 8 and 9. The porosity of two cores respectively increased from 19.4% to 19.7% and from 20.8% to 21.2%. The incremental extent of high permeability core was larger than that of low permeability core. The incremental tendency was more obvious for the permeability. The permeability of low permeability core increased from 3.66mD to 4.27mD respectively, rising by 16.8%. And the high permeability core increased from 16.28mD to 20.48mD, rising by 25.8%. The reason was with the permeability increasing, the overall radius of the pore throat presented an increasing tendency. It could increase the contact area between acidic fluids and rocks. Compare to the permeability reduction caused by the particle migration blockage, permeability increasing caused by the dissolution was dominated in this process. It ultimately resulted in an incremental in permeability.

The NMR curves in Fig. 8(b) and 9(b) both exhibited a distinct bimodal structure. The significant increment in the right-peak once again indicated that the positive impact of the dissolution reaction on permeability enhancement surpassed the negative impact of particle blockage. It was agreement with the permeability measurement results.

In this section, experiments were carried out to measure the porosity and permeability changes of three core samples with different permeabilities. The porosity of three cores showed an increasing tendency. In NMR T_2 curves, the porosity enhancement was reflected by an increasing in signal intensity. The total signal amplitude after the reaction was higher than that before the reaction for all three cores. With an increase in initial permeability, the incremental tendency of porosity also increased. The reason is a larger initial permeability corresponded to a larger average pore-throat radius. It could provide a large reaction space for the CO_2 -water-rock interactions. As a result, the dissolution effect was more pronounced in cores with higher permeability, which also lead to a larger increment in the porosity.

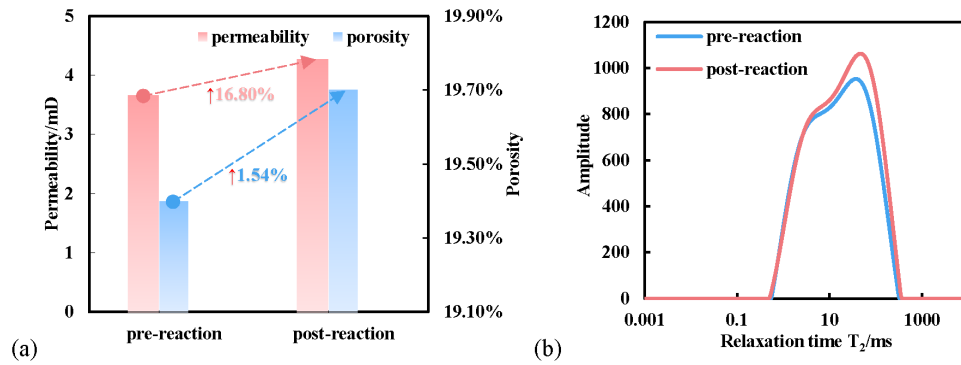


Figure 8—Properties change of a core with initial gas permeability of 31.2mD: (a) porosity and permeability; (b) NMR T_2 spectra

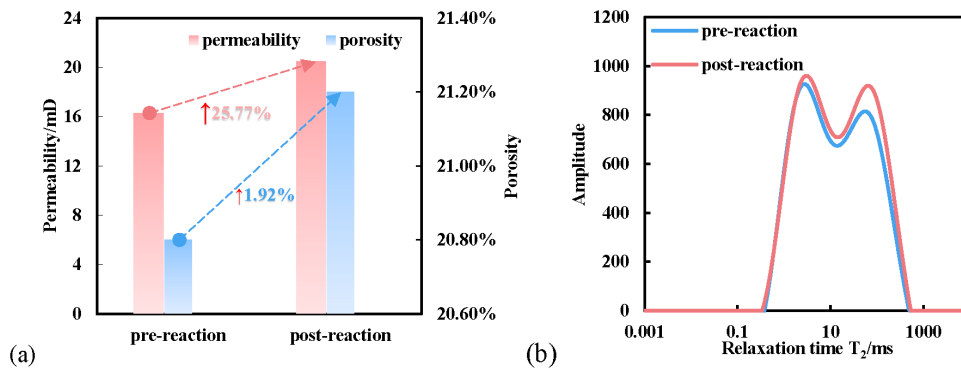
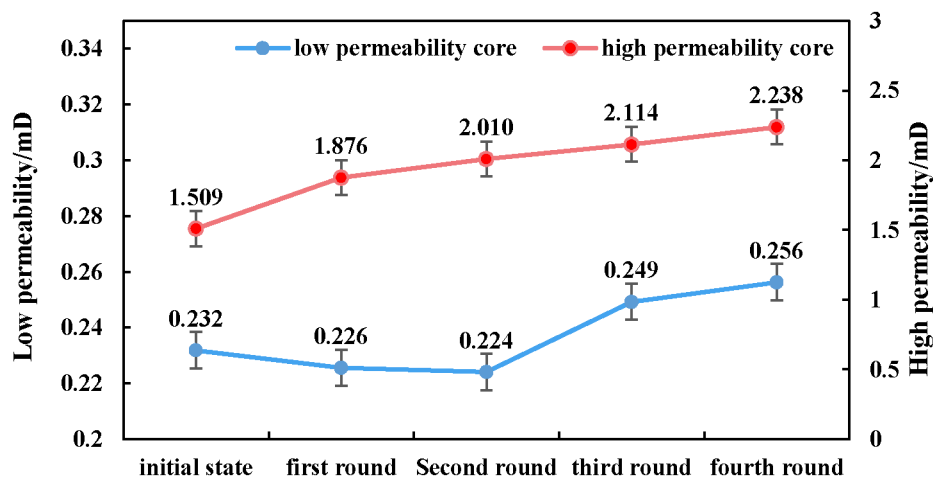


Figure 9—Properties change of a core with initial gas permeability of 53.7mD: (a) porosity and permeability; (b) NMR T_2 spectra

Variation Laws of Physical Properties in Heterogeneous Reservoirs

Permeability Variation Analysis. The permeability variation trend curves of two cores during four cycles of CO_2 injection were presented in Fig. 10. The increment of the permeability of the high-permeability core was similar in each round of CO_2 injection. It increased from an initial value of 1.509 mD to 2.238 mD, presenting an increase in the range of 48.31%. It can be attributed to the larger pore throat radius of the high-permeability core, which mitigated the detrimental effects of inorganic salt precipitation and particle transport on the reservoir. Consequently, the physical property changes of the high-permeability cores were predominantly influenced by the mineral dissolution. In contrast, the permeability change of the low-permeability core was more intricate. It initially decreased and subsequently increased. Its overall trend aligns with that of the high-permeability core, as permeability increased from an initial value of 0.232 mD to 0.256 mD, corresponding to an 8.7% of increment. The reason was the low-permeability core was more susceptible to the influence of inorganic salt precipitation. Under the dual influence of inorganic salt precipitation and mineral migration, the permeability reduced in the early-stage. With the round of injection cycles increase, the precipitates dissolve, and the mineral dissolution effect intensified. It will lead to the formation of dissolution pores and micro-cracks and then enhanced the permeability. With the increase of CO_2 injection cycles, the permeability of both cores generally increased. However, the increment was more pronounced for the high-permeability cores. It resulted in the heterogeneity enlargement, and the brine measurement permeability difference of dual cores increased from 6.5 to 8.7.

Figure 10—Permeability change curves during CO₂ multiple cycles injection

Nuclear Magnetic Resonance Results Analysis. NMR T_2 spectra can reflect the pore - throat radius distribution changing. As shown in Fig. 11, the NMR T_2 spectrum of the high - permeability core shifts right after multiple injection cycles, indicating that CO₂ dissolution increased the pore - throat radius. The more incremental signal intensity in the first two cycles meant the dissolution occurred in the early stage. The reason was as displacement proceeding, clay particles and soluble minerals decreased, and the remaining pore - surface minerals were more stable and less prone to dissolution. The NMR curve present a clear bimodal pattern, dividing pores into the large and small categories. The large - pore signal intensity increased more than that of small pores, which meant the strong CO₂ dissolution occurred in the large pores. The reason for that was CO₂ was hard to enter the small pores, falling to timely supply of ions like CO₃²⁻ and H⁺ that were required for the water-rock reactions. It was not conducive to the progress of the dissolution reaction. Moreover, particles released from the large-pore dissolution may be trapped in the small pores and reduced their NMR signals. These effects resulted in few changes for the small-pore signal intensity.

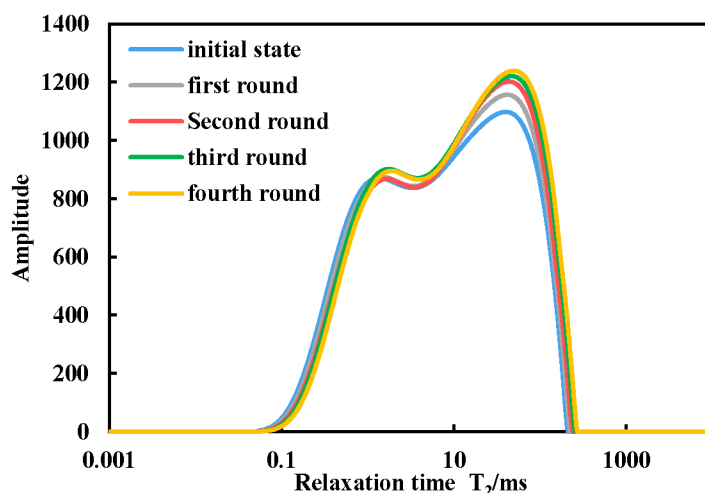
Figure 11—NMR T_2 spectrum curves of high-permeability cores

Fig. 12 presented the NMR T_2 signal amplitude of the low-permeability core initially declined in the early stage due to quick inorganic salt precipitation upon CO₂-formation water contact. It will slow down the mineral dissolution. The T_2 signal variation amplitude of low-permeability core was smaller than that in the high-permeability core after reaction. The reason was the pore throat size of the low-permeability core was smaller, and the contact area between the reaction fluid and the minerals was also smaller, which

lead to a slow progress of the water-rock reaction. With displacement proceeding, inorganic salt precipitates dissolved and the erosion pores further developed. Therefore, the NMR T_2 signal amplitude increased to a certain extent in the last two rounds.

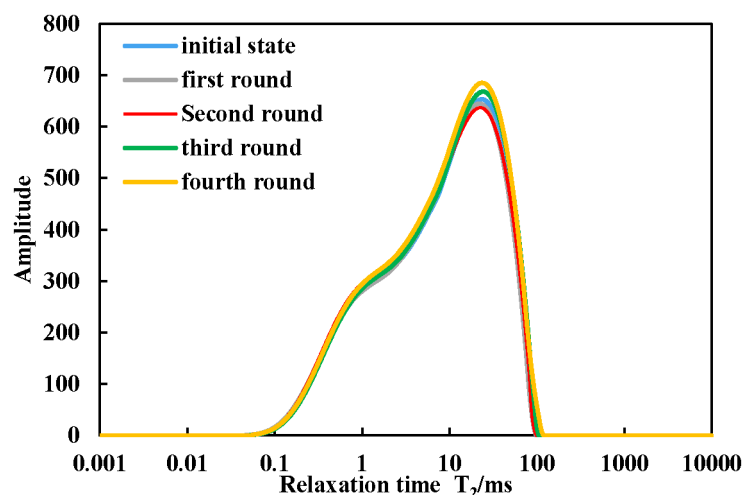


Figure 12—NMR T_2 spectrum curves of low-permeability cores

Conclusions

This work performed the inorganic salt precipitation, mineral dissolution, multi-stage CO_2 injection experiments under reservoir conditions of 110°C and 30 MPa. It clarified the variation laws and characteristics of physical properties in heterogeneous low - permeability sandstone reservoirs caused by CO_2 -brine-rock interactions. The following conclusions can be drawn from the experimental data:

1. Inorganic salt precipitation mainly involved carbonate minerals. Precipitates uniformly coated pore and throat surfaces, reducing their radii. However, the impact of inorganic salt precipitation on the uniformity of pore and throat sizes was relatively minor.
2. SEM images reveal vugs and micro-fractures caused by intense mineral dissolution, as well as clay-particle migration and accumulation on pore surfaces. The NMR T_2 signal amplitude of large pores changed more in the NMR curves, which indicated that dissolution mainly occurred in the large pores.
3. In the dual-core multiple round CO_2 injection experiments, the permeability of high-permeability core increased from 1.509 mD to 2.238 mD, while that of low -permeability core only increased from 0.232 mD to 0.256 mD. Consequently, the permeability ratio rose from 6.5 to 8.7. It indicated the long-term CO_2 injection could intensify the intra and inter-layer heterogeneity in the low-permeability sandstone reservoirs.

Acknowledgements

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